

$$\beta = \xi \smile \xi, \quad \mathcal{L}_{\text{res}} = \mathbb{C}\beta \subset B^2, \quad \mathcal{P}_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}})/Z^1$$

$$\bar{d} : \mathcal{P}_{\text{res}} \xrightarrow{\sim} \mathcal{L}_{\text{res}} \text{ removes the primitive choice.}$$

$$\text{Full Zariski tangent space: } T = H^1(C^\bullet) = \mathbb{C}\alpha \oplus \mathbb{C}\xi$$

$$\text{Response domain: } T_{\text{resp}} = \mathbb{C}\alpha$$

$$[\alpha, Z^1] = 0, \quad [\xi, \xi] = 2\beta \neq 0$$

$$Z = Z^1, \quad V_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}}), \quad \pi : C^2 \rightarrow C^2/\mathcal{L}_{\text{res}}$$

$$\mathfrak{o} : T \rightarrow \text{Hom}(Z, C^2) \oplus \text{Hom}(V_{\text{res}}, C^2/\mathcal{L}_{\text{res}})$$

$$a \mapsto (D_a|_Z, \pi D_a|_{V_{\text{res}}}), \quad T_{\text{resp}} = \ker \mathfrak{o}$$

$$\Theta(a) = \bar{D}_a \bar{d}^{-1} = \rho_{\text{res}}(a) \text{id}_{\mathcal{L}_{\text{res}}} \quad (a \in T_{\text{resp}}), \quad \rho_{\text{res}}(\alpha) = 2$$

$$\text{If } a = s\alpha + t\xi \text{ with } t \neq 0, \text{ then } D_a\xi = 2t\beta.$$

$$\text{Descent of } QD_a \text{ through } V_{\text{res}}/Z \text{ forces } Q(\beta) = 0.$$

Thus output quotienting cannot extend this response to a transverse direction while retaining a nonzero image of the marked resonant line.